

Regex vs Python re — Benchmark Results

Ratio = Python time ÷ Regex time. >1x = Regex faster.

OS	macOS-26.5.2-arm64-arm-64bit-Mach-O
Python	3.14.6
CPU	Apple M3 Pro
RAM	36 GB
Mojo	Mojo 1.0.0 (ed45d567)

Benchmark	Regex (µs)	Python (µs)	Ratio
Throughput scaling			
throughput_literal_100B	0.005	0.085	16.4x
throughput_literal_10KB	0.106	2.613	24.6x
throughput_literal_100KB	0.972	25.094	25.8x
throughput_literal_1MB	10.228	248.022	24.2x
throughput_class_10KB	0.308	61.035	197.9x
throughput_nomatch_100KB	0.966	24.811	25.7x
Anchors			
anchorbol	0.000	0.187	412.1x
anchoreol	0.006	0.197	33.9x
anchorwordboundary	0.005	0.250	50.0x
anchorwordboundarymiss	0.009	0.277	30.4x
anchorbolmiss_10KB	0.001	0.070	87.6x
Multiline / DOTALL			
multilinebol_findall_100_lines	1.287	12.794	9.9x
multilineeol_findall_100_lines	7.075	51.880	7.3x
dotall_multiline_body	0.002	0.092	38.4x
Named groups			
named_group_date	0.004	0.107	26.7x
named_group_email	0.009	0.124	13.3x
positional_group_email	0.011	0.124	11.5x
Negative lookahead			
neg_lookahead	0.005	0.220	44.0x
neg_lookbehind	0.004	0.215	52.4x
password_validation_lookahead	0.022	0.306	13.7x
Alternation			
alternation_4	0.002	0.087	48.3x
alternation_16	0.001	0.099	99.1x
alternation_16_miss	0.001	0.079	158.8x
Findall			
findall_3_matches	0.168	0.190	1.1x

Benchmark	Regex (µs)	Python (µs)	Ratio
findall_100_matches	1.230	9.281	7.5x
findall_500_dot_matches	2.418	13.977	5.8x
Replace			
replace_50_matches	1.002	5.470	5.5x
replace_named_backref	0.038	0.277	7.2x
Split			
split_100_parts	1.033	6.966	6.7x
Pathological patterns			
pathological_optional_16	0.010	1087.194	113249.4x
pathological_dotstar_anchored_5K	0.122	1.454	11.9x
pathological_dotstar_miss_5K	0.001	3.462	3147.0x
pathological_triple_backref	0.008	0.096	11.7x
pathological_nested_quantifier_miss	8.334	38.416	4.6x
Real-world patterns			
realworld_url_parse	0.017	0.402	23.8x
realworld_phone	0.006	0.252	44.2x
realworld_hex_color	0.003	0.190	61.3x
realworld_semver	0.013	0.363	27.7x
realworld_key_value_findall	0.251	0.626	2.5x
realworld_html_tag_findall	0.291	0.656	2.3x
realworld_ws_normalize	0.104	0.610	5.9x
realworld_log_search_1000_lines	0.821	11.009	13.4x
Inline flags			
inline_ignorecase	0.004	0.192	47.9x
inline_multiline_search	0.006	0.278	44.2x
Engine comparison			
engine_dfa_no_capture	0.004	0.224	53.3x
engine_pike_with_capture	0.007	0.222	33.6x
engine_backtrack_with_backref	0.008	0.206	25.4x
Regex faster: 47 slower: 0			